

On Analogies of Dynamical Systems

At a fundamental level, no system is truly static and nothing happens for no reason. The nature of a dynamical system is that it changes - for some reason. The study of dynamical systems seeks to describe the behavior of systems as they change. As one might imagine, there are a multitude of factors and influences which affect the behavior of various systems. Of specific interest is that which encourages or discourages a system to change. What motivations or hesitations, attractions or repulsions influence its behavior? Are these influences governed by the nature of the system itself or the environment in which the system evolves? Or both? If so, which has a dominant effect? This is a question of “how?”, not of “why?”

It is not merely that change is interesting, but that it is essential to life and survival - it is the driving force. When surveying for a target or prey, it is optimal for a predator to scan the scene with their eyes rather than staring in one direction. Focusing attention in one location creates a central bias which tends to ignore the periphery - where change is more likely to occur. Scanning attention across locations creates a peripheral bias where change is more likely to occur, thus ignoring the central image which is less detrimental to spotting motion (change). When confined to a static environment of solitude and bleakness, the human mind will begin to hallucinate as it seeks a sign of change - something that says there is still life in the system.

The nature of change is fundamental to physics and philosophy. All physical laws depend on a description of change (or lack thereof), and there a wide variety of fields and practices in which the patterns and structures of dynamical systems can be identified and applied. Dynamical systems provides a rigorous mathematical framework to describe these changes, and applying the formalism to other disciplines could yield curious insights and applications.

Of key significance is the classical relationship between mass, acceleration, and force - described by Newton’s Second Law - and the relationship between charge, the electric field, and the electrostatic force - given by Coulomb’s Law:

Newton’s Second Law:

$$F = ma$$

Coulomb’s Law:

$$F = qE$$

Note both have similar forms, each denoting a force as the product of two interacting components - a source (mass, m , or charge, q) and a field (acceleration, a , or electric field E). Interestingly enough, one can combine the words “source” and “field” to create the word “force”. This also makes for a nice mnemonic. So, the generalized force can be expressed as: $Force = Source \times Field$

Also note that force is measured in Newtons in both cases. Mass is measured in kilograms and charge is measured in Coulombs. Thus, the units of acceleration are Newtons per kilogram, or meter per second squared, and the units of the electric field are Newtons per Coulomb, or

Volts per meter (remember this for later). If the electric field is considered as analogous to an acceleration field, then the unit for generalized acceleration is given by: $\text{Acceleration} = \frac{\text{Force}}{\text{Source}}$

In essence, Sources interact with Fields and experience Forces, and Forces are the product of an interaction between a Source and a Field.

This generalization can be recognized in Lewin's equation to describe the behavior of a person in their environment (citation): $B = f(P, E)$

which is generally stated as: $\text{Behavior} = \text{Person} \times \text{Environment}$

Here, the units of each parameter are more qualitative than quantitative, though the formalism is easily identified.

Sources carry intrinsic information such as mass, charge, energy, even genetics. Fields carry extrinsic information about the environment. Forces inevitably carry information about both intrinsic and extrinsic properties of the system. This means that it is not possible to immediately infer whether a force or outcome is due solely to intrinsic or extrinsic properties without a controlled experimental variance of either property to test for such influence.

It is curious to consider the possibility that Fields could be influenced by Sources themselves. Perhaps the motion of the Source through the Field had a lasting effect on it, perhaps dissipating Field Energy or something of the sort - like walking through snow and leaving behind a trail of deformation. It's worth noting that this may have something to do with the nature of the Field - as the Field itself has a Source! So perhaps a Source can have a lasting impact on another. Consider too that this may relate to the nature of conservative forces and nonconservative forces and the fields through which they act.*

To explore the nature of this interaction, consider two masses placed a distance, R , apart from each other. Assuming they are isolated from the influence of any other source of mass, the force experienced by each mass is the same as described by Newton's Third Law, and the force between them is described by Newton's Universal Law of Gravitation:

$$\text{Force} = G \frac{mM}{R^2}$$

where G is the universal gravitational constant, M is the larger mass - sometimes called the parent or planet, and m is the smaller mass - the child or satellite.

Now consider Coulomb's Law, $\text{Force} = k \frac{qQ}{R^2}$

where the source terms are now charges q and Q , and the constant k is the Coulomb constant. Note the similarities between these two equations.

To isolate the effect of the Field on one of the Sources (say, the satellite), simply rearrange the equation $\text{Force} = G \frac{mM}{R^2} \frac{\text{Force}}{m} = G \frac{M}{R^2}$ and since the acceleration is the ratio of force per mass by Newton's Second Law, $a = G \frac{M}{R^2}$

This says that the acceleration experienced by m is caused by M . So, from the perspective of m , the properties of the acceleration field are extrinsic and caused by M . The force that m experiences in the field also depends on its own intrinsic properties - specifically mass. The more massive m becomes, the slower it will accelerate provided the same field at the same location. For now, it is assumed that m cannot influence the field produced by M .

Since Forces are the product of Sources and Fields, they are influenced by both intrinsic and extrinsic properties.

Potential Energy and Potential

The configuration of a Source in a Field, and its corresponding Force, can also be described via its location within the Field. This description is known as Potential Energy, and can be thought of as a sort of “spatial energy”.

Gravitational Potential Energy, $U = mgh = Fh$

where the gravitational force by Newton’s Second Law is $F = mg$.

Note that the potential energy is a product of Force and Distance, and thus has units of $N \cdot m$ which is a unit of Energy measured in Joules.

The distance h is the distance from the field’s zero-potential or “ground”.

A similar expression can be seen in the Electrostatic Potential Energy, where $U = qEs = Fs$

where the electrostatic force is given by $F = qE$.

If either a mass or a charge (or a mass with charge) is “elevated” to a non-zero potential, there is energy which “wishes” to be released. Upon releasing the mass or charge, or setting it on its way with some impetus, it will be driven by a force to follow a trajectory which seeks to minimize this potential energy.

The force which develops as a result of this potential energy minimization can be described independently of the source (mass/charge), more specifically, as the gradient of the potential function.

$$F = -\nabla\phi$$

Define potential as the ratio of potential energy per source (specific potential energy? energy density?) such that it is the product of Field and (Reference) Displacement.

For electric potential, commonly known as voltage, $V = \frac{U}{q} = Es$

has units of Joules per Coulomb or Volts.

For gravitational potential, not commonly known as gravitational voltage, $V = \frac{U}{m} = gh$

which has units of Joules per kilogram or $\frac{m^2}{s^2}$. Note these are the same units as velocity-squared.

Work and Kinetic Energy

Moving the source of mass/charge through the field requires energy - specifically at the expense of potential energy. Energy involved in moving the source from one potential to another is known as Work and is defined as

$$W = Fd\cos\theta$$

where W is measured in Joules [J], F is the force measured in Newtons [N], d is the distance in meters [m], and θ is the angle between the applied force and the line of action (direction of motion).

Angles greater than zero indicate suboptimal force influence on the motion of the source/object. This is more accurately described in the dot-product relationship of work as

$$W = \int \vec{F} \cdot d\vec{r}$$

since work is the energy which contributes to change in motion along a line of action (given by the vector $\vec{r}$), this also defines Kinetic Energy.

$$W = \Delta KE$$

The change in motion is the result of the net work done by the system, from both conservative forces efficiently transferring energy and nonconservative forces inefficiently transferring (removing) energy. When energy is fully conserved, all potential energy is converted into kinetic energy. However, in real systems, energy is lost in one form or another to nonconservative forces such as friction.

The work done by these nonconservative force is described using the original definition,

$$W_{nc} = \int \vec{F}_{nc} \cdot d\vec{r}$$

While nonconservative forces impact the kinetic energy of the system, conservative forces and the work they perform depend only on (the loss of) potential energy.

$$W_c = \int \vec{F}_c \cdot d\vec{r} = -\Delta PE$$

The conservation of energy statement fully describes this exchange.

$$KE_i + PE_i + W_{nc} = KE_f + PE_f$$

Thus, for sake of showing consistency in the definitions...

The total work is the sum of the conservative and nonconservative work:

$$KE_i + PE_i - PE_f + W_{nc} = KE_f$$

$$KE_i + -(PE_f - PE_i) + W_{nc} = KE_f$$

$$-\Delta PE + W_{nc} = KE_f - KE_i$$

$$W_c + W_{nc} = \Delta KE = W$$

And the change in energy of the system is zero when energy is conserved.

$$-\Delta PE + W_{nc} = KE_f - KE_i$$

$$W_{nc} = \Delta KE + \Delta PE$$

which says that energy is either added or removed from the system (not constant). If all of the work is conservative and the nonconservative work is zero,

$$0 = \Delta KE + \Delta PE$$

and energy is conserved (remains constant).

Dynamics

If the energy of a system can be fully described, then its behavior can be described too.

The energy of a system can be described with three energy modes:

- **Potential energy:** energy stored in *configuration* (position, deformation, separation).
- **Kinetic energy:** energy stored in *motion*.
- **Dissipative energy:** energy *lost* to nonconservative effects (friction, drag, resistance).

Potential energy represents the energy associated with a configuration in a field, while kinetic energy represents the energy associated with movement through a field. Dissipative energy is energy lost to nonconservative effects like friction, drag, and electrical resistance.

Each of these three forms of energy represents a passive role in the system.

- **store energy in a potential-like way**
- **store energy in a kinetic-like way**
- **or dissipate energy**

RLC Circuit

In a classical RLC circuit, the three passive components are the resistor, inductor, and capacitor. The resistor is the dissipative element. The inductor is the kinetic storage element (in the magnetic field, and in the inertia-like behavior of current). The capacitor is the potential storage element (in the electric field, and in charge separation).

Before the circuit is closed, the capacitor is uncharged—there is no voltage across it—and the inductor carries no current. When the switch closes, current starts to flow. The resistor opposes the flow and converts electrical power into heat. The inductor opposes changes in the flow, so it “pushes back” against sudden jumps in current. The capacitor begins accumulating charge, which builds a voltage across it.

As the capacitor charges, the voltage across it rises, leaving less voltage available to drive current through the loop. The current falls. In the ideal DC limit, current goes to zero once the capacitor reaches the source voltage. So the circuit evolves from kinetic activity (current) into potential storage (charge separation), while the resistor continuously bleeds energy away.

Mass-Spring-Damper

A similar behavior appears in mechanics with the mass-spring-damper system.

In a classical mass-spring-damper system, the three passive components are the damper, mass, and spring. The damper is the dissipative element: it converts mechanical power into heat through friction-like effects. The mass is the kinetic storage element: it stores energy in motion and resists changes in velocity. The spring is the potential storage element: it stores energy in deformation and “pushes back” when stretched or compressed.

Before anything moves, the spring can be undeformed (no spring force) and the mass can be at rest (no kinetic energy). When the system is disturbed—pulled, released, or driven—motion begins. The damper opposes motion and dissipates energy, producing a resistive force that grows with velocity. The mass resists rapid changes in velocity, so it “pushes back” against sudden acceleration. The spring begins to stretch or compress, building a restoring force as it stores potential energy.

As the spring deformation increases, the restoring force grows, reducing the net force available to keep accelerating the mass. The velocity tends to peak and then fall as the spring pulls the mass back toward equilibrium. In the absence of continued driving, the damper steadily bleeds energy out of the motion, so the oscillations shrink and the system settles toward rest. So the mechanical system evolves through an exchange between kinetic energy (mass in motion) and potential energy (spring deformation), while the damper continuously drains energy away.

This is not just a poetic similarity — the governing relationships line up in the same roles:

- The mass resists changes in velocity (a mechanical “inductor”).

- The spring produces a restoring force proportional to displacement (a mechanical “capacitor,” in the sense of potential storage).
- The damper produces a resistive force proportional to velocity (a mechanical “resistor”).

In other words, the components line up by role:

Resistor (R) Damper (b): dissipates energy and drains motion/flow into heat.

Inductor (L) Mass (m): stores kinetic energy and resists changes in flow/velocity.

Capacitor (C) Spring Compliance (1/k): stores potential energy and builds a restoring “push” as it charges/deforms. (Stiffness k is the inverse role.)

Constitutive Laws

The mapping above becomes concrete when you write the element laws in matching form.

Electrical (one common form): - Resistor: $V_R = RI$ - Inductor: $V_L = L\dot{I}$ - Capacitor: $I_C = C\dot{V}$

Mechanical (translation, with $v = \dot{x}$): - Damper: $F_d = bv$ - Mass: $F_m = m\dot{v}$ ($= m\ddot{x}$) - Spring: $F_s = kx$ (equivalently, $\dot{F}_s = kv$)

That one line you wrote down — $F = m\dot{v}$ — is the mechanical twin of the inductor law $V = L\dot{I}$.

Both say: the kinetic storage element resists *changes in the flow variable*.

Restriction–Inertance–Accumulator (Hydraulics)

A similar behavior appears in hydraulics with a restriction–inertance–accumulator system.

In a classical hydraulic loop, the three passive components can be taken as: a restriction (a valve/orifice or narrow section of pipe), an inertance (a long pipe / moving slug of fluid), and a compliance (an accumulator with a flexible diaphragm or trapped compressible volume). The restriction is the dissipative element: it converts mechanical power into heat through viscous losses and turbulence. The inertance is the kinetic storage element: the moving fluid has inertia and resists rapid changes in flow (the same effect behind water hammer). The compliance is the potential storage element: it stores energy by compressing/expanding a volume (diaphragm stretch, trapped gas compression, etc.).

Before anything happens, the fluid can be at rest (no kinetic energy), and the accumulator can be relaxed (no stored pressure energy). When a pump is engaged or a valve is opened to a pressurized reservoir, flow begins. The restriction immediately “pushes back” against that flow by demanding a pressure drop to sustain it. The inertance also pushes back, but differently: it resists sudden changes in flow, so the flow cannot jump instantly. Meanwhile, the accumulator

begins to accept fluid, which stretches/compresses its compliant element and builds pressure as it stores potential energy.

As the accumulator pressurizes, its back-pressure rises, leaving less pressure difference available to keep driving flow through the loop. The flow rate peaks and then falls as the pressure levels approach equilibrium. In the steady limit (no oscillatory driving), flow can drop toward zero even though the system remains pressurized—because the compliance has saturated or “charged,” just like a capacitor. Throughout the entire process, the restriction steadily bleeds energy away.

So all three systems share the same basic structure: two elements exchange energy back and forth (kinetic potential), while the dissipative element continuously bleeds energy away. The nouns change—charge and current versus displacement and velocity—but the behavior is the same: storage, exchange, decay.

— AI —

A useful way to see why these analogies keep working is to start from a more general lens that already appears in the earlier work: **potential is energy per source**. Gravitational potential is energy per unit mass, and electrical potential is energy per unit charge. Once potential is treated as “energy per source,” the idea of a *through* quantity becomes natural: it is a **source-flow rate** (charge flow, mass flow, volume flow, etc.). The earlier notes make this explicit by introducing a gravitational “Ohm’s law” where **B is mass flow rate** (mass-current) and the proportionality constant is a **flow resistance** (“Gohs”).

That same reasoning also produces a clean capacitance-style result: if **H** represents a *mass-capacitance*, then the target form is “source proportional to potential,” directly mirroring ($q = CV$).

With that framing, the terminal pairs (voltage/current, force/velocity, pressure/flow, potential/mass-flow) stop looking arbitrary. They are the pairs that make power bookkeeping consistent.

Terminal pairs and the three passive roles

Domain / Port	Driving variable (effort)	Through variable (flow)	Power
Electrical	V	I	$P = VI$
Mechanical (translation)	F	v	$P = Fv$
Mechanical (rotation)	τ	ω	$P = \tau\omega$
Hydraulic	Δp	Q	$P = \Delta p Q$
Gravitational (transport port)	V_g (J/kg)	$B = \dot{m}$ (kg/s)	$P = V_g B$

— AI —

At the terminals of a system, there is usually a “driving difference” and a resulting “through” quantity, corresponding to how the potential and kinetic elements exchange energy through the system.

In many systems, the ratio of driving difference to through quantity is set by the dissipative path, and it controls how quickly energy is bled away relative to how much is stored. This is known as **impedance**, and it is defined for each system as the ratio of “pressure” to “flux”:

- Electrical: $Z = \frac{V}{I}$
- Mechanical (translation): $Z = \frac{F}{v}$
- Mechanical (rotation): $Z = \frac{\tau}{\omega}$
- Hydraulic: $Z = \frac{\Delta p}{Q}$

Note that this is not a ratio of stored energies. It’s a ratio of the **driving variable** to the **through variable** at a port/terminal.

— AI — For a purely dissipative path, that ratio is just the resistive constant. Once storage is present, the “effective ratio” becomes dynamic (time / frequency dependent), because part of the driving difference is temporarily in storage before being returned or dissipated. — AI —

It now becomes natural to describe systems using paired variables like voltage/current, force/velocity, and pressure/flow — because these are the parameter pairs that describe how energy propagates through the system.

Power Accounting

Because these paired variables describe how energy propagates through the system, their product conveys a different meaning: **power**, the rate at which energy is transferred.

- Electrical: $P = VI$
- Mechanical (translation): $P = Fv$
- Mechanical (rotation): $P = \tau\omega$
- Hydraulic: $P = \Delta p Q$

Describing a system at its terminals using (V, I) or (F, v) or $(\Delta p, Q)$ makes it possible to directly track **energy transfer**.

This also clarifies why real sources “sag under load.” Real sources contain internal dissipative effects, so part of the driving difference is spent internally when the through quantity increases.

A battery is the clean electrical example: it behaves like an EMF in series with an internal resistance, so the terminal voltage decreases as current increases:

$$V_{\text{terminal}} = \mathcal{E} - rI$$

A pressurized canister behaves the same way in hydraulics: the reservoir pressure is the available driving difference, but the outlet pressure falls as flow increases because pressure is lost across internal restrictions. Same structure, different nouns.

At this point it becomes reasonable to give these paired variables a generic name. Bond-graph prior art calls them **effort** (the driving difference) and **flow** (the through quantity), precisely because their product is power and because the same three passive roles (potential storage, kinetic storage, dissipation) can be written consistently in those variables.

This is the bridge to dynamical systems. The moment a system includes storage — something that accumulates over time — its behavior stops being a purely algebraic “ratio at the terminal” and becomes an evolution law. That is where ODEs (and, when the storage is distributed in space, PDEs) enter the conversation.

Before discussing differential equations, a table is presented which collects and organizes the analogy thusfar.

Component / parameter (catalog row)	Gravita- tional (potential / mass-flow port)	Electrical	Mechanical (transla- tion)	Mechanical (rotation)	Hydraulic
Displace- ment / coordinate	Height / position h	Plate separation / path coordinate x (contextual)	Position x	Angle θ	Volume- coordinate / compliance volume V
Motion = displace- ment/time	Mass-flow speed is contextual; port flow is $B = \dot{m}$	Carrier drift speed u_d (contextual)	Velocity $v = \dot{x}$	Angular speed $\omega = \dot{\theta}$	Volume flow rate $Q = \dot{V}$
Source (field- domain “stuff”)	Gravitational Mass m	Electrical Charge q	—	—	—

Component / parameter (catalog row)	Gravita- tional (potential / mass-flow port)	Electrical	Mechanical (transla- tion)	Mechanical (rotation)	Hydraulic
Field = Force/Source	Gravitic field g	Electric field E	Acceleration a	Angular accel. α	Pressure- gradient / force-density view (contextual)
Force = Source · Field	$F = mg$	$F = qE$	$F = ma$	$\tau = J\alpha$	$F = pA$ (local relation)
Work- Energy = Force · Dis- placement	$U = mgh$	$U = \int F dx$ (e.g., qEs)	$U = \int F dx$	$U = \int \tau d\theta$	$U = \int p dV$
Potential = En- ergy/Source	Gravitational potential $V_g = U/m = gh$	Voltage $V = U/q$	(Not typically used as “per mass” here)	(Not typically used as “per mass” here)	Pressure $p = U/V$ (energy density)
Pressure	—	—	—	—	p (Pa = N/m ² = J/m ³)
Flux / current	Mass flow $B = \dot{m}$	Current $I = \dot{q}$	—	—	$Q = \dot{V}$ (volume flux)
Current density / flux density	Mass flux density (contextual)	Current density J (A/m ²)	—	—	Volumetric flux density q_v (m/s) (contextual)
Port “driving difference” (bond- graph effort <i>e</i>)	Potential V_g	Voltage V	Force F	Torque τ	Pressure difference Δp
Port “through quantity” (bond- graph flow <i>f</i>)	Mass flow B	Current I	Velocity v	Angular speed ω	Volume flow Q

Component / parameter (catalog row)	Gravita- tional (potential / mass-flow port)	Electrical	Mechanical (transla- tion)	Mechanical (rotation)	Hydraulic
Power at a port	$P = V_g B$	$P = VI$	$P = Fv$	$P = \tau\omega$	$P = \Delta p Q$
Impedance (port ratio)	$Z_g = V_g/B$	$Z = V/I$	$Z = F/v$	$Z = \tau/\omega$	$Z = \Delta p/Q$
Resistance (R-type coefficient)	R_g with $V_g = R_g B$ (your “Gohs”-style slot)	R with $V = RI$	b with $F = bv$	b_θ with $\tau = b_\theta\omega$	R_h with $\Delta p = R_h Q$
Conduc- tance / mobility (G-type coefficient)	$G_g = 1/R_g$ with $B = G_g V_g$	$G = 1/R$ with $I = GV$	$\mu = 1/b$ with $v = \mu F$	$\mu_\theta = 1/b_\theta$ with $\omega = \mu_\theta \tau$	$G_h = 1/R_h$ with $Q = G_h \Delta p$
Capaci- tance / compliance (C-type coefficient)	Mass- capacitance H with $m = HV_g$ and $B = H\dot{V}_g$	C with $q = CV$ and $I = C\dot{V}$	$C_m = 1/k$ with $x = C_m F$	$C_\theta = 1/k_\theta$ with $\theta = C_\theta \tau$	C_h with $V = C_h \Delta p$ and $Q = C_h \Delta \dot{p}$
Stiffness / elastance (inverse of C)	$1/H$	$1/C$	k	k_θ	$E_h = 1/C_h$
Inductance / inertance (I-type coefficient)	L_g with $\pi_g = L_g B$ and $V_g = L_g \dot{B}$	L with $\lambda = LI$ and $V = L\dot{I}$	Mass m with $p = mv$ and $F = m\dot{v}$	Inertia J with $L = J\omega$ and $\tau = J\dot{\omega}$	Inertance I_h with $\Pi = I_h Q$ and $\Delta p = I_h \dot{Q}$
C-state variable (bond- graph $q = \int f dt$)	Mass m	Charge q	Displacement x	Angle θ	Volume V
I-state variable (bond- graph $p = \int e dt$)	Potential- impulse $\pi_g = \int V_g dt$ (formal slot)	Flux linkage λ	Momentum p	Angular momentum L	Pressure impulse Π

Component / parameter (catalog row)	Gravita- tional (potential / mass-flow port)	Electrical	Mechanical (transla- tion)	Mechanical (rotation)	Hydraulic
Unit signature noted in the doc (catalog tags)	Capaci- tance: [Source/Po- tential]; Conduc- tance: [Flow/Poten- tial]; Inductance: [Area/Source]	same tag-structure applied by role	same tag-structure applied by role	same tag-structure applied by role	same tag-structure applied by role

make sure all table parameters are properly dervied/defined

relate to generalized bond-graph systems

analogy between effort/flow and drive/response to lead into dynamical systems/oscilla-
tors/waves